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United States Patent Application Publication

20250279498

Kind Code

A1

Publication Date

September 04, 2025

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BATTERY AND POWER CONSUMING APPARATUS WITH SAME

Abstract

A battery includes a box body, and multiple battery cells and a heat conducting member arranged in the box body. Each battery cell is provided with multiple side walls. The multiple side walls include a first side wall and a second side wall connected to each other. The first side wall is a side wall with the largest area of the battery cell. The second side walls of at least part of the battery cells are connected to the box body in a heat conducting manner through the heat conducting member.

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Appl. No.: 19/212755

Filed: May 20, 2025

Foreign Application Priority Data

CN

202321086507.4

May. 08, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2023/132383 20231117 PENDING child US 19212755

Publication Classification

Int. Cl.: H01M10/653 (20140101); H01M10/658 (20140101); H01M50/204 (20210101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of PCT Application PCT/CN2023/132383, filed Nov. 17, 2023, which claims priority to Chinese Patent Application No. 202321086507.4, filed with the China National Intellectual Property Administration on May 8, 2023 and entitled “BATTERY AND POWER CONSUMING APPARATUS WITH SAME”, each are incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of batteries, and in particular, to a battery and a power consuming apparatus with the same.

BACKGROUND

[0003] A battery generally includes a box body and battery cells arranged in the box body.

[0004] For the battery cells, unexpected temperature increases may occur due to overheating, short circuits, overcharging, self-heating, or mechanical collisions. If the heat generated during the temperature increase exceeds the speed of heat dissipation to the environment and the temperature continues to increase, when certain critical temperature is reached, especially when the temperature at which the separator of the battery collapses is reached, the battery cells may be broken down and thermal runaway may be caused. In the existing batteries, when one of the battery cells experiences thermal runaway, it is easy to cause thermal diffusion throughout the battery, ultimately leading to a safety accident of the battery and reducing the use safety of the battery.

[0005] Therefore, there is an urgent need for a battery that can meet the requirement of no thermal diffusion.

SUMMARY

[0006] In view of this, according to a first aspect of the present application, a battery is provided, which can meet the requirement of no thermal diffusion to guarantee the use safety of the battery.

[0007] The battery according to the embodiment of the present application includes: a box body, multiple battery cells, and a heat conducting member. The multiple battery cells are arranged in the box body. Each battery cell is provided with multiple side walls. The multiple side walls include a first side wall and a second side wall. The first side wall and the second side wall are connected to each other. The first side wall is a side wall with the largest area of the battery cell. The heat conducting member is arranged in the box body. The second side walls of at least part of the battery cells are connected to the box body in a heat conducting manner through the heat conducting member.

[0008] In the battery according to the embodiment of the present application, by connecting the second side walls of at least part of the battery cells to the box body in a heat conducting manner through the heat conducting member, heat generated by the battery cells can be directly transferred to the box body through the heat conducting member, thus preventing a large amount of heat from being transferred to adjacent battery cells when thermal runaway occurs in one of the battery cells, making the battery meet the requirement of no thermal diffusion, and improving the safety performance of the battery.

[0009] Optionally, the heat conducting member is connected to different side walls of the box body in a heat conducting manner. A contact area between the heat conducting member and the box body can be increased, thus further ensuring that the heat conducting member can effectively transfer the heat on the battery cells to the box body, and ensuring that the battery can meet the requirement of

no thermal diffusion.

[0010] In some embodiments, a first heat conducting adhesive layer is provided between the heat conducting member and the box body. While achieving a fixed connection between the heat conducting member and the box body, it can also be ensured that the heat conducting member can effectively transfer the heat on the battery cells to the box body.

[0011] In some embodiments, multiple rows of battery units are provided in the box body. Each row of the battery units includes multiple battery cells arranged along a first direction, and the heat conducting member is provided between the second side walls of at least one row of the battery units and the box body in the first direction. It can be ensured that when thermal runaway occurs in at least one row of the battery cells, the heat can be directly transferred to the box body, thus preventing this row of the battery cells from transferring a large amount of heat to adjacent rows of battery cells, so as to achieve the purpose of preventing thermal diffusion.

[0012] In some embodiments, at least two rows of the battery units are arranged in a staggered manner in the first direction so that the first side walls of adjacent rows of battery cells arranged opposite each other are staggered in the first direction. A contact area between adjacent rows of two battery cells arranged opposite each other can be reduced, thus reducing heat transfer between the two battery cells arranged opposite each other, preventing thermal diffusion, and ensuring that the battery can meet the requirement of no thermal diffusion.

[0013] In some embodiments, orthographic projections of the first side walls of the adjacent rows of battery cells arranged in a staggered manner in a second direction have non-overlapping portions, and the non-overlapping portions are in contact with the heat conducting member in a heat insulating manner. Contact in a heat insulating manner can further reduce heat transfer between two battery cells which are arranged opposite each other, and ensure that the battery can meet the requirement of no thermal diffusion.

[0014] In some embodiments, the heat conducting member includes a heat conducting layer and a heat insulating layer which are arranged in a stacked manner. The heat conducting layer is connected to the second side wall in a heat conducting manner. The heat insulating layer is in contact with the non-overlapping portions. By connecting the heat conducting layer with the second side wall in a heat conducting manner, the heat generated by the battery cells can be directly transferred to the box body through the heat conducting member, thus preventing a large amount of heat from being directly transferred to adjacent battery cells. The heat insulating layer is in contact with the non-overlapping portions to achieve heat insulating match between the battery cells arranged opposite each other, thus reducing the probability of heat transfer between the two battery cells arranged opposite each other, further preventing a large amount of heat from being transferred to other battery cells when thermal runaway occurs in the battery cells, and making the battery meet the requirement of no thermal diffusion.

[0015] In some embodiments, a second heat conducting adhesive layer is provided between the heat conducting layer and the opposite second side wall. While achieving a fixed connection between the heat conducting member and the battery cell, it can also be ensured that the heat on the battery cell can be effectively transferred to the heat conducting member, thus making it easier to transfer the heat on the battery cell to the box body through the heat conducting member to achieve the purpose of heat dissipation.

[0016] In some embodiments, a thickness of the heat conducting layer is greater than that of the heat insulating layer. Therefore, the heat conducting performance of the heat conducting member can be ensured, thus ensuring that the heat generated by the battery cell when thermal runaway occurs can be effectively transferred to the box body through the heat conducting member.

[0017] In some embodiments, a ratio of the thickness of the heat insulating layer to the thickness of the heat conducting layer is not greater than $\frac{1}{2}$. Therefore, the heat conducting performance of the heat conducting member can be further ensured, thus ensuring that the heat generated by the battery cell when thermal runaway occurs can be effectively transferred to the box body through the heat

conducting member.

[0018] In some embodiments, the heat conducting member is filled in a gap between the battery cells and an inner wall of the box body. While ensuring that a volume of the battery is not increased, the heat conducting member can be effectively arranged between the battery cells and the box body to transfer the heat on the battery cells to the box body through the heat conducting member. In addition, by using the heat conducting member, position stability between the battery cells and the box body can be improved.

[0019] In some embodiments, the heat conducting member is constructed to be deformable to buffer a stress. That is to say, the heat conducting member has a buffering effect, thus protecting the battery through the heat conducting member, improving the squeezing and impact resistance of the battery, and extending the service life of the battery.

[0020] In a second aspect, the present application provides a power consuming apparatus, which includes a battery, and the battery is the battery described in the above embodiment. The battery is configured to provide electric energy.

[0021] In the technical solution of the embodiment of the present application, by adopting the battery described in the above embodiment, the use safety of the power consuming apparatus can be improved, and the service life of the power consuming apparatus can be extended.

[0022] Additional aspects and advantages of the present application will be provided in the following description, some of which will become apparent from the following description or may be learned from practices of the present application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The foregoing and/or additional aspects and advantages of the present application will become apparent and comprehensible from the following descriptions of embodiments with reference to the accompanying drawings, where:

[0024] FIG. 1 is a schematic diagram of a battery according to some embodiments of the present application;

[0025] FIG. 2 is a top view of a battery according to some embodiments of the present application;

[0026] FIG. 3 is a partial enlarged view of area I in FIG. 2;

[0027] FIG. 4 is a schematic diagram of a heat conducting member according to some embodiments of the present application; and

[0028] FIG. 5 is a schematic structural diagram of a power consuming apparatus according to some embodiments of the present application.

REFERENCE NUMERALS

[0029] **1000**—power consuming apparatus; **200**—battery; **300**—controller; **400**—motor; [0030]

10A—battery unit; **10**—battery cell; **11**—side wall; **111**—first side wall; **112**—second side wall;

[0031] **20**—non-overlapping portion; [0032] **30**—box body; [0033] **40**—heat conducting member;

41—heat conducting layer; **42**—heat insulating layer; [0034] X—first direction; Y—second direction; and Z—third direction.

DETAILED DESCRIPTION

[0035] To make the purposes, technical solutions and advantages of the embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be clearly described below with reference to the drawings in the embodiments of the present application. Clearly, the described embodiments are merely some rather than all of the embodiments of the present application. All other embodiments obtained by those of ordinary skill in the art based on the embodiments of the present application without making any inventive labor still fall within the scope of protection of the present application.

[0036] Unless otherwise defined, meanings of all technical and scientific terms used in the present application are the same as those usually understood by those skilled in the art to which the present application belongs. In the present application, terms used in the description of the present application are merely intended to describe objectives of specific embodiments, but are not intended to limit the present application. The terms “include”, “comprise”, and any variant thereof in the description and the claims of the present application and the description of the drawings are intended to cover a non-exclusive inclusion. Terms such as “first” and “second” in the description, claims and accompanying drawings of the present application are used for distinguishing different objects, instead of describing a specific order or primary and secondary relationships.

[0037] “Embodiment” mentioned in the present application means that particular features, structures, or characteristics described with reference to the embodiment may be included in at least one embodiment of the present application. The term appearing at different positions of the description may not necessarily refer to the same embodiment or an independent or alternative embodiment that is mutually exclusive with another embodiment. A person skilled in the art explicitly or implicitly understands that the embodiments described herein may be combined with other embodiments.

[0038] In the description of the present application, it should be noted that unless otherwise explicitly specified and defined, the terms “mount”, “connect”, “connected”, and “attach” should be understood in a broad sense. For example, it may be fixed connection, detachable connection, or integral connection; or it may be direct connection, an indirect connection through an intermediate medium, or internal communication between two components. For those of ordinary skill in the art, the specific meanings of the above terms in the present application may be interpreted according to specific situations.

[0039] In the present application, the term “and/or” is merely an association to describe associated objects, and means that three relationships may exist. For example, A and/or B may represent the following three cases: only A exists, both A and B exist, and only B exists. In addition, the character “/” in the present application generally indicates an “or” relationship between associated objects.

[0040] In the embodiments of the present application, the same reference numerals represent the same components, and for the sake of simplicity, detailed descriptions of the same components are omitted in different embodiments. It should be understood that the thickness, length, width, and other dimensions of various components in the embodiments of the present application shown in the accompanying drawings, as well as the overall thickness, length, width, and other dimensions of the integrated apparatus, are only exemplarily descriptive and should not constitute any limitation on the present application.

[0041] At present, from the development of the market situation, the application of batteries **200** is becoming increasingly widespread. Batteries **200** are not only applied to energy storage power source systems such as water, fire, wind and solar power stations, but also widely applied to electric transport tools, such as electric bicycles, electric motorcycles, and electric vehicles, as well as many fields, such as military equipment and aerospace.

[0042] With the continuous expansion of the application field of the batteries **200**, the market demand is also constantly expanding.

[0043] A battery **200** generally includes structures such as a box body **30** and battery cells **10**. In a working process of the battery **200**, the battery cells **10** may experience unexpected temperature increases due to overheating, short circuits, overcharging, self-heating, or mechanical collisions occurring in the battery cells **10**. If the heat generated during the temperature increase exceeds the speed of heat dissipation to the environment and the temperature continues to increase, when the continuously increasing temperature reaches certain critical temperature, especially when the temperature at which a separator of the battery **200** collapses is reached, the battery cells **10** may be broken down and thermal runaway may be caused.

[0044] The applicant has noticed that in the existing battery **200**, when thermal runaway occurs in one of the battery cells **10** of the battery **200**, heat may be quickly transferred to adjacent battery cells **10**, causing thermal diffusion throughout the battery **200**, resulting in low safety of the battery **200** and shortening the service life of the battery **200**.

[0045] To solve the above problems, an embodiment of the present application provides a battery **200** that can reduce rapid diffusion of heat between adjacent battery cells **10**. The specific solution is to arrange a heat conducting member **40** in a box body **30**, and connect second side walls **112** of at least part of the battery cells **10** to the box body **30** in a heat conducting manner through the heat conducting member **40**. In this way, when thermal runaway occurs in the battery cells **10**, the heat conducting member **40** can directly transfer the heat on the battery cells **10** to the box body **30**, thus preventing a large amount of heat from being directly transferred to the adjacent battery cells **10** when thermal runaway occurs in one of the battery cells **10**, that is, preventing a large amount of heat from being rapidly transferred between adjacent battery cells **10**, so as to improve the safety performance of the battery **200** and extend the service life of the battery **200**.

[0046] It should be noted that the battery **200** disclosed in the embodiment of the present application may be applied to, but not limited to, a power consuming apparatus **1000** such as a vehicle, a ship or an air vehicle. A power supply system of the power consuming apparatus **1000** may be composed of the battery **200** disclosed in the present application, which is beneficial for improving the use safety of the power supply system in the power consuming apparatus **1000** and extending the service life of the power supply system.

[0047] The battery **200** according to the embodiment of the present application will be described below with reference to FIG. 1 to FIG. 4.

[0048] As shown in FIG. 1 and FIG. 2, the battery **200** according to the embodiment of the present application includes: a box body **30**, multiple battery cells **10**, and a heat conducting member **40**.

[0049] As shown in FIG. 1 and FIG. 2, the multiple battery cells **10** are arranged in the box body **30**. The battery cells **10** can be supported and protected by using the box body **30**, thus extending the service life of the battery cells **10** and improving the position stability of the battery cells **10**.

[0050] In some examples, an accommodating cavity is formed in the box body **30**, and the multiple battery cells **10** are all arranged in the accommodating cavity to achieve the purpose of arranging the multiple battery cells **10** in the box body **30**.

[0051] As shown in FIG. 2 and FIG. 3, each battery cell **10** is provided with multiple side walls **11**. The multiple side walls **11** include a first side wall **111** and a second side wall **112**. The first side wall **111** and the second side wall **112** are connected to each other. The first side wall **111** is a side wall with the largest area of the battery cell **10**. Here, it may also be understood that each battery cell **10** includes a first side wall **111** and a second side wall **112** which are connected to each other. An area of the second side wall **112** is smaller than that of the first side wall **111**, so that the first side wall **111** is formed as the side wall with the largest area of the battery cell **10**.

[0052] As shown in FIG. 2 and FIG. 3, the heat conducting member **40** is arranged in the box body **30**. The second side walls **112** of at least part of the battery cells **10** are connected to the box body **30** in a heat conducting manner through the heat conducting member **40**. That is to say, at least part of the battery cells **10** are connected to the box body **30** in a heat conducting manner through the heat conducting member **40**. In this way, while achieving fixed connection between at least part of the battery cells **10** and the box body **30**, it is easy to transfer the heat on the battery cells **10** to the box body **30** through the heat conducting member **40**, thus preventing the energy released by the thermal runaway of the battery cells **10** in the battery **200** from being all transferred to the adjacent battery cells **10**, preventing thermal runaway between the adjacent battery cells **10** and improving the use safety of the battery **200**.

[0053] From the above structure, it can be seen that in the battery **200** in the present application, the heat conducting member **40** is arranged in the box body **30**, and the second side walls **112** of at least part of the battery cells **10** are connected to the box body **30** in a heat conducting manner

through the heat conducting member **40** to achieve fixed connection between at least part of the battery cells **10** and the box body **30**, so that at least part of the battery cells **10** can be stably arranged in the box body **30**, thus improving the position stability of at least part of the battery cells **10**, that is, improving the structural stability of the battery **200**, and ensuring the working performance of the battery **200**.

[0054] At the same time, because the second side walls **112** of at least part of the battery cells **10** are connected to the box body **30** in a heat conducting manner through the heat conducting member **40**, when thermal runaway occurs in the battery cells **10**, the heat generated on the battery cells **10** can be transferred to the box body **30** in time through the heat conducting member **40** to prevent a large amount of heat on the battery cells **10** from being directly transferred between two adjacent battery cells **10**, thus preventing thermal runaway from occurring between the adjacent battery cells **10**, making the battery **200** meet the requirement of no thermal diffusion, and improving the safety performance of the battery **200**.

[0055] In addition, by connecting the second side walls **112** of at least part of the battery cells **10** to the box body **30** in a heat conducting manner through the heat conducting member **40**, the heat generated by the battery cells **10** during working can be directly conducted out through the box body **30**, thus improving the heat dissipation and cooling effect of the battery cells **10**.

[0056] That is to say, in the present application, by arranging the heat conducting member **40** that achieves the heat conducting connection between at least part of the battery cells **10** and the box body **30**, while avoiding thermal runaway between the adjacent battery cells **10**, the heat dissipation effect of the battery cells **10** can be improved, thus reducing the possibility of thermal runaway occurring in the battery cells **10**.

[0057] It may be understood that, compared with the prior art, in the present application, the second side walls **112** of at least part of the battery cells **10** are connected to the box body **30** in a heat conducting manner through the heat conducting member **40**, so that the heat on the battery cells **10** can be transferred to the box body **30** in time, thus preventing a large amount of heat from being directly transferred between two adjacent battery cells **10**, that is, preventing thermal runaway between the adjacent battery cells **10**, making the battery **200** meet the requirement of no thermal diffusion, and improving the safety performance of the battery **200**.

[0058] In some embodiments, the heat conducting member **40** is connected to different side walls of the box body **30** in a heat conducting manner. Here, it refers to that the box body **30** is provided with multiple different side walls, and the heat conducting member **40** is simultaneously connected to the different side walls of the box body **30** to increase a contact area between the heat conducting member **40** and the box body **30**, thus ensuring that the heat conducting member **40** can effectively transfer the heat on the battery cells **10** to the box body **30**, making the battery **200** meet the requirement of no thermal diffusion and improving the use safety of the battery **200**.

[0059] In addition, by connecting the heat conducting member **40** to the different side walls of the box body **30** in a heat conducting manner, the connection strength between the heat conducting member **40** and the box body **30** can be also ensured, so as to use the box body **30** to support the heat conducting member **40** and improve the structural stability of the heat conducting member **40**, thus ensuring that the heat conducting member **40** can effectively transfer the heat on the battery cells **10** to the box body **30**.

[0060] In some examples, the box body **30** is provided with a bottom wall and end walls connected to the bottom wall. The bottom wall and the end walls cooperate to form an accommodating cavity. The multiple battery cells **10** and the heat conducting member **40** are all arranged in the accommodating cavity, and the heat conducting member **40** is simultaneously connected to the bottom wall and the end walls of the box body **30** to achieve the purpose of connecting the heat conducting member **40** to different side walls of the box body **30** in a heat conducting manner, thus ensuring that the heat conducting member **40** can transfer the heat on the battery cells **10** to the end walls and the bottom wall of the box body **30** in time when thermal runaway occurs in the battery

cells **10**.

[0061] Optionally, an extension height of the heat conducting member **40** in a third direction Z is adapted to an extension height of the box body **30** in the third direction Z. The third direction Z mentioned here may be understood as a Z direction shown in FIG. **1**, i.e., a height direction of the box body **30**. Through the above arrangement, when the heat conducting member **40** is connected to the end walls of the box body **30** in a heat conducting manner, the contact area between the heat conducting member **40** and the box body **30** can be further increased, so that the heat on the battery cells **10** can be transferred to the box body **30** in time, thus preventing the heat from being diffused between the battery cells **10**.

[0062] In some embodiments, a first heat conducting adhesive layer is provided between the heat conducting member **40** and the box body **30**. The first heat conducting adhesive layer is used for achieving heat conducting connection between the heat conducting member **40** and the box body **30**, thus ensuring that the heat conducting member **40** can transfer the heat on the battery cells **10** to the box body **30** in time, that is, the first heat conducting adhesive layer can transfer the heat on the battery cells **10** to the box body **30** in time, so that the heat on the battery cells **10** in which thermal runaway occurs can be dissipated in time.

[0063] At the same time, by arranging the first heat conducting adhesive layer between the heat conducting member **40** and the box body **30**, the first heat conducting adhesive layer can also be used for achieving fixed connection between the heat conducting member **40** and the box body **30**, so that a position of the heat conducting member **40** relative to the box body **30** is stable, thus improving the working performance of the heat conducting member **40**.

[0064] In specific examples, the first heat conducting adhesive layer is provided between the heat conducting member **40** and the end walls and the bottom wall of the box body **30**, so that the heat conducting member **40** can be connected to different side walls of the box body **30** in a heat conducting manner, thus improving the connection strength between the heat conducting member **40** and the box body **30**, while ensuring the heat conducting effect between the heat conducting member **40** and the box body **30**.

[0065] Optionally, the first heat conducting adhesive layer is formed of acrylic, heat conducting silicone, and the like. While ensuring that the first heat conducting adhesive layer has a heat conducting effect, it can also ensure that the first heat conducting adhesive layer can effectively achieve fixed connection between the heat conducting member **40** and the box body **30**.

[0066] Optionally, a thickness value range of the first heat conducting adhesive layer is 2 mm-5 mm. When the thickness of the first heat conducting adhesive layer is less than 2 mm, the heat conducting effect and bonding effect of the first heat conducting adhesive layer may be reduced. When the thickness of the first heat conducting adhesive layer is greater than 5 mm, on the one hand, the manufacturing costs of the first heat conducting adhesive layer may be increased, and on the other hand, the space occupied by the first heat conducting adhesive layer may also be increased, so that when the first heat conducting adhesive layer is arranged in the box body **30**, the space occupied by the battery **200** may be increased.

[0067] Therefore, in the present application, the thickness value range of the first heat conducting adhesive layer is set to 2 mm-5 mm, thus reducing the manufacturing costs of the first heat conducting adhesive layer, reducing the space occupied by the first heat conducting adhesive layer, and lowering the assembling difficulty of the first heat conducting adhesive layer, while ensuring the heat conducting effect and bonding effect of the first heat conducting adhesive layer.

[0068] In specific examples, the thickness of the first heat conducting adhesive layer may be 2 mm, 3 mm, 4 mm, 5 mm, or the like.

[0069] Optionally, a second heat conducting adhesive layer is provided between the heat conducting member **40** and the opposite second side wall **112**. That is to say, the heat conducting adhesive layer is not only provided between the heat conducting member **40** and the box body **30**, but also between the heat conducting member **40** and the opposite second side wall **112**. The

second heat conducting adhesive layer is used for achieving heat conducting connection between the heat conducting member **40** and the battery cells **10**, thus ensuring that the heat on the battery cells **10** can be transferred to the heat conducting member **40** in time. Then, the heat conducting member **40** transfers the received heat on the battery cells **10** to the box body **30**, thus achieving the purpose of transferring the heat on the battery cells **10** to the box body **30** in time, so that the heat on the battery cells **10** in which thermal runaway occurs can be dissipated in time and the heat can be prevented from being diffused.

[0070] For the material, thickness and the like of the second heat conducting adhesive layer, refer to the first heat conducting adhesive layer, which will not be repeated here.

[0071] In some embodiments, as shown in FIG. 1 and FIG. 2, multiple rows of battery units **10A** are provided in the box body **30**. Each row of the battery units **10A** includes multiple battery cells **10**. The multiple battery cells **10** are arranged along a first direction X. The heat conducting member **40** is provided between the second side walls **112** of at least one row of the battery units **10A** and the box body **30** in the first direction X. The first direction X mentioned here may be understood as an X direction shown in FIG. 1 and FIG. 2, i.e., is a length direction of the box body **30**. That is to say, multiple rows of battery units **10A** are arranged in the box body **30** in the present application, and each row of the battery units **10A** includes multiple battery cells **10** arranged along the length direction of the box body **30**.

[0072] Through the above arrangement, the space inside the box body **30** can be properly utilized to ensure that the battery **200** can include multiple battery cells **10**, thus increasing the capacity of the battery **200**. At the same time, it can also reduce the difficulty in connecting the multiple battery cells **10** and the difficulty in connecting the multiple rows of battery units **10A**, thus improving the assembling efficiency of the battery units **10A**.

[0073] At the same time, the heat conducting member **40** is provided between the second side walls **112** of at least one row of the battery units **10A** and the box body **30**, so as to transfer the heat on at least one row of the battery units **10A** to the box body **30** through the heat conducting member **40**, thus preventing a large amount of heat from being diffused between adjacent rows of battery units **10A**, and preventing heat from being directly transferred between adjacent rows of battery cells **10**, so as to avoid thermal diffusion and make the battery **200** meet the requirement of no thermal diffusion.

[0074] In some examples, the heat conducting member **40** is provided between the second side walls **112** of the battery cells **10** located at end portions of a row of battery units **10A** and the box body **30**. That is to say, in a row of battery units **10A**, the second side walls **112** of the battery cells **10** located at the end portions of the battery units **10A** are connected to the box body **30** through the heat conducting member **40**, so as to transfer heat on the battery units **10A** to the box body **30**.

[0075] Through the above arrangement, when thermal runaway occurs in the battery units **10A**, the heat can be transferred between adjacent battery cells **10** in the battery units **10A** and to the battery cells **10** located at the end portions of the battery units **10A**. After the heat is transferred to the battery cells **10** at the end portions, the heat is transferred to the box body **30** through the heat conducting member **40**, thus achieving the purpose of transferring the heat on the battery units **10A** to the box body **30** through the heat conducting member **40**, preventing the heat from being transferred between adjacent battery units **10A**, and preventing the heat from being diffused in the box body **30**.

[0076] In specific examples, as shown in FIG. 1 and FIG. 2, the heat conducting member **40** is provided between the second side walls **112** of the battery cells **10** at the end portions of each row of the battery units **10A** and the box body **30**. That is to say, the end portions of each row of the battery units **10A** are connected to the box body **30** through the heat conducting member **40**, thus ensuring that the heat on each row of the battery units **10A** can be transferred to the box body **30** in time, and improving the use safety of the battery **200**.

[0077] It should be further noted that since the heat conducting member **40** is provided between the

second side wall **112** of the battery cell **10** and the box body **30**, the heat conducting member **40** can be fixed by using a pre-tightening force between the battery cell **10** and the box body **30**, thus improving the position stability of the heat conducting member **40**, ensuring that the heat conducting member **40** can transfer the heat, and further improving the structural stability of the battery **200**.

[0078] In some examples, the thickness of the heat conducting member **40** is equal to a distance between the second side wall **112** of the battery cell **10** and the box body **30**. Here, the thickness of the heat conducting member **40** may be understood as a dimension of the heat conducting member **40** extending along the first direction X. By arranging the heat conducting member **40** between the second side wall **112** of the battery cell **10** and the box body **30**, the heat conducting member **40** can be fixed by using the pre-tightening force between the battery cell **10** and the box body **30**, thus improving the position stability of the heat conducting member **40**.

[0079] In some embodiments, as shown in FIG. 1 and FIG. 2, at least two rows of the battery units **10A** are arranged in a staggered manner in the first direction X so that the first side walls **111** of adjacent rows of battery cells **10** arranged opposite each other are staggered in the first direction X. That is to say, at least two rows of the battery units **10A** are arranged in a staggered manner along the length direction of the box body **30**, so that the first side walls **111** of adjacent rows of battery cells **10** arranged opposite each other are staggered in the first direction X. In this way, a contact area between two battery cells **10** arranged opposite each other in adjacent rows can be reduced, thus reducing heat transfer between the two battery cells **10** arranged opposite each other, i.e., preventing the heat on one of the battery cells **10** from being all transferred to the opposite battery cell **10A** when thermal runaway occurs in the battery cell **10**, so as to prevent thermal diffusion and ensure that the battery **200** can meet the requirement of no thermal diffusion.

[0080] At the same time, the above arrangement can also enable one battery cell **10** in two adjacent rows of battery units **10A** to be arranged directly facing towards two battery cells **10**, so as to transfer the heat on the battery cell **10** to different battery cells **10**, achieve the purpose of dispersing the heat, and prevent excessive heat from being transferred to one battery cell **10** to cause thermal runaway in the battery cell **10**, thus improving the use safety of the battery cells **10**.

[0081] Optionally, as shown in FIG. 1 and FIG. 2, multiple rows of battery units **10A** are arranged in the box body **30**, and every two adjacent rows of battery units **10A** are arranged in a staggered manner in the first direction X, so that the first side walls **111** of the adjacent rows of battery cells **10** arranged opposite each other are staggered in the first direction X, thus avoiding thermal diffusion between adjacent rows of battery units **10A** and improving the use safety of the battery cells **10**.

[0082] Optionally, in adjacent rows of battery cells **10** arranged opposite each other, the contact area between the two battery cells **10** arranged opposite each other is $\frac{1}{3}$ to $\frac{1}{2}$ of the area of the first side wall **111**. In this way, when thermal runaway occurs in one battery cell **10**, the temperature transferred to the adjacent battery cells **10** can be reduced by half, thus reducing the heat transfer between adjacent battery cells **10**, avoiding thermal diffusion, and ensuring that the battery **200** can meet the requirement of no thermal diffusion.

[0083] In some embodiments, as shown in FIG. 1, FIG. 2 and FIG. 3, orthographic projections of the first side walls **111** of the adjacent rows of battery cells **10** arranged in a staggered manner in a second direction Y have non-overlapping portions **20**, and the non-overlapping portions **20** are in contact with the heat conducting member **40** in a heat insulating manner. The second direction Y mentioned here may be understood as a Y direction shown in FIG. 2, i.e., a width direction of the box body **30**. That is to say, when the first side walls **111** of adjacent rows of battery cells **10** arranged opposite each other are staggered in the first direction X, the orthographic projections of the first side walls **111** of the adjacent rows of battery cells **10** arranged in a staggered manner in the second direction Y have the non-overlapping portions **20**. The non-overlapping portions **20** are in contact with the heat conducting member **40** in a heat insulating manner to avoid direct heat

transfer between the battery cells **10**, that is, avoid thermal diffusion between the battery cells **10** through the heat conducting member **40**, thus ensuring the heat insulating effect between the battery cells **10** and ensuring that the battery **200** can meet the requirement of no heat diffusion. [0084] It should be noted that the non-overlapping portion **20** mentioned above may be understood as a part of the first side wall **111** of the battery cell **10** that is not in contact with the opposite battery cell **10**.

[0085] To sum up, it can be seen that the heat conducting member **40** in the present application can not only achieve a heat conducting effect, but also achieve a heat insulating effect, thus preventing the heat from being transferred between the battery cells **10** while ensuring that the heat on the battery cells **10** can be effectively transferred to the box body **30**.

[0086] In addition, by enabling the non-overlapping portions **20** to be in contact with the heat conducting member **40** in a heat insulating manner, it can also achieve the effect of properly utilizing the space inside the box body **30** to arrange the heat conducting member **40**, so as to avoid excessive increase in the volume of the battery **200** due to the arrangement of the heat conducting member **40**.

[0087] That is to say, in the present application, while ensuring that the heat conducting member **40** can effectively transfer the heat on the battery cells **10** to the box body **30** and avoid heat transfer between the battery cells **10** arranged opposite each other, it can also avoid excessive increase in the volume of the battery **200** due to the arrangement of the heat conducting member **40**, thus reducing the area occupied by the battery **200** and lowering the difficulty in arranging the battery **200**.

[0088] In some examples, two opposite sides of the heat conducting member **40** are in contact with two non-overlapping portions **20** in a heat insulating manner respectively, and a width of the heat conducting member **40** directly facing towards the non-overlapping portions **20** is equal to a distance between the two non-overlapping portions **20**. The width of the heat conducting member **40** directly facing towards the non-overlapping portions **20** mentioned here may be understood as the dimension of the heat conducting member **40** extending along the second direction Y, so as to nest the heat conducting member **40** directly facing towards the non-overlapping portions **20** between two battery cells **10**. In this way, the heat conducting member **40** can be conveniently fixed by using the pre-tightening force between the two battery cells **10** to improve the position stability of the heat conducting member **40**.

[0089] Optionally, a height of the heat conducting member **40** directly facing towards the non-overlapping portions **20** is equal to that of the battery cell **10**, so as to increase a contact area between the heat conducting member **40** and the battery cell **10** and improve the heat insulating effect of the heat conducting member **40**.

[0090] In some other examples, when the orthographic projections of the first side walls **111** of adjacent rows of battery cells **10** arranged in a staggered manner in the second direction Y have non-overlapping portions **20**, the battery **200** further includes a monitor arranged directly facing towards the non-overlapping portions **20**. The monitor is configured to monitor the performance of the battery cells **10** during working and improve the use safety of the battery cells **10**.

[0091] That is to say, in the present application, by properly arranging the positions of adjacent rows of battery cells **10**, space can be provided for arranging other structural members (such as the monitor), thus ensuring that the monitor can be arranged in the box body **30** to monitor the performance of the battery cells **10** during working, and also avoiding the increase in space occupied by the battery **200** due to the arrangement of the monitor.

[0092] It should be noted that since the multiple rows of battery units **10A** have multiple non-overlapping portions **20**, in the present application, part of the non-overlapping portions **20** are in contact with the heat conducting member **40** in a heat insulating manner, and the monitor is arranged directly facing towards the other part of the non-overlapping portions **20** to achieve the proper utilization of the space inside the box body **30**, thus not only preventing thermal diffusion

from occurring between the two battery cells **10** arranged opposite each other, but also facilitating the monitoring of the performance of the battery cells **10** during working.

[0093] Optionally, the monitor includes a pressure sensor box, a gas analysis sensor box, and a lithium plating monitoring instrument box. The monitor is configured to respectively monitor positive and negative voltages of each row of the battery cells **10** and the position of an explosion-proof valve, thus achieving detailed monitoring of the performance and changes of each battery cell **10**, and ensuring the use safety of the battery cells **10**.

[0094] Optionally, sizes of the pressure sensor box, the gas analysis sensor box, and the lithium plating monitoring instrument box match the distance between the two non-overlapping portions **20**, so that the monitor can be fixed by using the pre-tightening force between the two battery cells **10**, thus improving the position stability of the monitor and reducing the fixed connection of the monitor.

[0095] In some examples, the monitor is arranged between two battery cells **10** and welded to the box body **30**, so as to further fix the monitor and improve the position stability of the monitor.

[0096] Optionally, a buffer pad is arranged between the monitor and the battery cells **10**, and the buffer pad is wrapped and combined with the battery cells **10**. In this way, when the battery **200** is squeezed, the buffer pad can be triggered in time to provide buffer protection for the battery cells **10**, thus reducing the impact of external squeezing force on the battery cells **10** and extending the service life of the battery cells **10**.

[0097] Optionally, the buffer pad may be a rubber buffer pad or a polyurethane buffer pad to provide a buffering effect.

[0098] In some examples, when the battery **200** is squeezed, the monitor can also obtain a squeezing value. When it is determined that the squeezing value reaches a squeezing threshold, a control system of the battery **200** may actively disconnect a high-voltage protection device to improve the use safety of the battery **200**.

[0099] In some embodiments, as shown in FIG. **3** and FIG. **4**, the heat conducting member **40** includes a heat conducting layer **41** and a heat insulating layer **42**. The heat conducting layer **41** and the heat insulating layer **42** are arranged in a stacked manner. The heat conducting layer **41** is connected to the second side walls **112** in a heat conducting manner. The heat insulating layer **42** is in contact with the non-overlapping portions **20**. Therefore, it can achieve the heat conducting connection between the heat conducting member **40** and the battery cells **10**, achieve contact between the heat conducting member **40** and the non-overlapping portions **20**, and ensure that the heat conducting member **40** can not only achieve a heat conducting effect, but also achieve a heat insulating effect, thus preventing the heat from being transferred between the battery cells **10** and improving the use safety of the battery **200** while ensuring that the heat on the battery cells **10** can be effectively transferred to the box body **30**.

[0100] Specifically, when the heat conducting layer **41** is connected to the second side walls **112** in a heat conducting manner, the heat conducting connection between the heat conducting member **40** and the battery cells **10** can be achieved. In this way, the heat conducting layer **41** can directly transfer the heat generated by the battery cells **10** to the box body **30**, thus preventing the heat from being transferred to adjacent battery cells **10**, improving the heat dissipation effect of the battery cells **10**, preventing the heat from being transferred between the battery cells **10**, and making the battery **200** meet the requirement of no heat diffusion. The heat insulating layer **42** is in contact with the non-overlapping portions **20** to achieve heat insulating match between the heat conducting member **40** and the non-overlapping portions **20**, thus achieving heat insulating match between the battery cells **10** arranged opposite each other, reducing the probability of heat transfer between the two battery cells **10** arranged opposite each other, further preventing the heat from being transferred to other battery cells **10** when thermal runaway occurs in the battery cells **10**, and making the battery **200** meet the requirement of no thermal diffusion.

[0101] In some examples, the heat conducting layer **41** is connected to the second side walls **112** in

a heat conducting manner and the heat conducting layer **41** is connected to the box body **30** in a heat conducting manner, so as to achieve the heat conducting connection between the second side walls **112** and the box body **30** through the heat conducting layer **41**.

[0102] Optionally, the heat conducting layer **41** is made of a material with a high thermal conductivity, and the thermal conductivity is greater than 1.2 W/m.K, so that the heat conducting layer **41** can effectively absorb the thermal runaway heat of the battery cells **10** and transfer the thermal runaway heat to the box body **30**.

[0103] Specifically, the heat conducting layer **41** may be made of a material with a relatively high thermal conductivity, such as a heat conducting phase-change material, a heat conducting graphite sheet, a heat conducting metal sheet, a heat conducting adhesive, or a heat conducting filler.

[0104] Optionally, the heat insulating layer **42** is made of a material with a low thermal conductivity, and the thermal conductivity is less than 0.1 W/m.K, so as to ensure that the heat insulating layer **42** can prevent thermal diffusion between the battery cells **10** arranged on two opposite sides of the heat insulating layer **42**, prevent thermal runaway from occurring between adjacent battery cells **10**, and thus make the battery **200** meet the requirement of no thermal diffusion.

[0105] Specifically, the heat insulating layer **42** may be formed as an aerogel heat insulating pad, a ceramic heat insulating pad, or the like.

[0106] Optionally, a thickness value range of the heat insulating layer **42** is 2 mm-4 mm. When the thickness of the heat insulating layer **42** is less than 2 mm, the heat insulating effect of the heat insulating layer **42** may be reduced. When the thickness of the heat insulating layer **42** is greater than 4 mm, on the one hand, the manufacturing costs of the heat insulating layer **42** may be increased, and on the other hand, due to the fixed thickness of the heat conducting member **40**, when the thickness of the heat insulating layer **42** is large, correspondingly the thickness of the heat conducting layer **41** may decrease subsequently, and the heat conducting performance of the heat conducting member **40** may be reduced.

[0107] Therefore, in the present application, by setting the thickness value range of the heat insulating layer **42** to 2 mm-4 mm, the manufacturing costs of the heat insulating layer **42** can be reduced and the heat conducting performance of the heat conducting member **40** can be guaranteed, while ensuring the heat insulating effect of the heat insulating layer **42**.

[0108] In specific examples, the thickness of the heat insulating layer **42** may be 2 mm, 3 mm, 4 mm, or the like.

[0109] In some embodiments, a second heat conducting adhesive layer is provided between the heat conducting layer **41** and the opposite second side wall **112**. That is to say, when the heat conducting member **40** includes the heat conducting layer **41**, the second heat conducting adhesive layer is provided between the heat conducting layer **41** and the opposite second side wall **112** to achieve the heat conducting connection between the heat conducting layer **41** and the opposite second side wall **112**, that is, to achieve the fixed connection between the heat conducting member **40** and the battery cells **10**. In this way, it can ensure that the heat on the battery cells **10** can be effectively transferred to the heat conducting member **40**, thus achieving the effect of transferring the heat on the battery cells **10** to the box body **30** through the heat conducting member **40**, achieving the purpose of heat dissipation, avoiding heat transfer between the battery cells **10**, and making the battery **200** meet the requirement of no heat diffusion.

[0110] In some embodiments, as shown in FIG. 4, the thickness of the heat conducting layer **41** is greater than that of the heat insulating layer **42**. Therefore, the heat conducting performance of the heat conducting member **40** can be ensured, thus ensuring that the heat generated by the battery cell **10** when thermal runaway occurs can be effectively transferred to the box body **30** in time through the heat conducting member **40**.

[0111] In some embodiments, a ratio of the thickness of the heat insulating layer **42** to the thickness of the heat conducting layer **41** is not greater than $\frac{1}{2}$. That is to say, the ratio of the thickness of the

heat insulating layer **42** to the thickness of the heat conducting layer **41** is less than or equal to $\frac{1}{2}$, so as to further ensure that the thickness of the heat conducting layer **41** is greater than that of the heat insulating layer **42**, make the thickness of the heat conducting layer **41** at least twice the thickness of the heat insulating layer **42**, thus ensure the heat conducting performance of the heat conducting member **40**, ensure that the heat conducting member **40** can effectively transfer the heat generated by the battery cells **10** when thermal runaway occurs to the box body **30**, and then guide the heat to the outside through the box body **30**, thus improving the use safety of the battery **200**.

[0112] In specific examples, the ratio of the thickness of the heat insulating layer **42** to the thickness of the heat conducting layer **41** is equal to $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, or the like.

[0113] In some embodiments, as shown in FIG. 1, FIG. 2 and FIG. 3, the heat conducting member **40** is filled in a gap between the battery cells **10** and an inner wall of the box body **30**. Therefore, the battery cells **10** can be connected to the box body **30** in a heat conducting manner through the heat conducting member **40**, thus improving the use safety of the battery **200**.

[0114] At the same time, the above arrangement can also avoid increasing the volume of the battery **200** due to the arrangement of the heat conducting member **40**, that is, avoid increasing the space occupied by the battery **200**, and reduce the difficulty in mounting the battery **200**.

[0115] In addition, filling the heat conducting member **40** in the gap between the battery cells **10** and the inner wall of the box body **30** can also limit the relative positions of the battery cells **10** and the box body **30** through the heat conducting member **40**, and prevent the battery cells **10** from shaking in the box body **30**, thus improving the position stability of the battery cells **10**, that is, improving the structural stability of the battery **200**.

[0116] To sum up, in the present application, by filling the heat conducting member **40** in the gap between the battery cells **10** and the inner wall of the box body **30**, while ensuring that the volume of the battery **200** is not increased, the heat conducting member **40** can be effectively arranged between the battery cells **10** and the box body **30**, so as to transfer the heat on the battery cells **10** to the box body **30** through the heat conducting member **40**. In addition, by using the heat conducting member **40**, the position stability between the battery cells **10** and the box body **30** can be improved.

[0117] In some embodiments, the heat conducting member **40** is constructed to be deformable to buffer a stress. That is to say, the heat conducting member **40** is configured to have a buffering effect, thus protecting the battery **200** through the heat conducting member **40**, improving the squeezing and impact resistance of the battery **200**, and extending the service life of the battery **200**.

[0118] In some examples, the heat conducting layer **41** and the heat insulating layer **42** of the heat conducting member **40** are both made of deformable materials. For example, the heat conducting layer **41** is made of a heat conducting adhesive, and the heat insulating layer **42** is made of an aerogel. In this way, while ensuring that the heat conducting member **40** has the functions of heat conduction and heat insulation, the heat conducting member **40** is deformed under external force, so as to protect the battery **200** through the heat conducting member **40**.

[0119] The battery **200** of the present application will be described below in detail with reference to the drawings.

[0120] As shown in FIG. 1 and FIG. 2, the battery **200** includes: a box body **30**, battery units **10A**, and a heat conducting member **40**.

[0121] As shown in FIG. 1 and FIG. 2, multiple rows of battery units **10A** are provided in the box body **30**. Each row of the battery units **10A** includes multiple battery cells **10**. The multiple battery cells **10** are arranged along a first direction X. Each battery cell **10** is provided with four side walls **11**. The four side walls **11** include first side walls **111** and second side walls **112**. The first side walls **111** and the second side walls **112** are connected to each other. The first side walls **111** are side walls with the largest area of the battery cell **10**.

[0122] As shown in FIG. 1 and FIG. 2, two adjacent rows of battery units **10A** are arranged in a

staggered manner in the first direction X so that the first side walls **111** of adjacent rows of battery cells **10** arranged opposite each other are staggered in the first direction X, thus making orthographic projections of the first side walls **111** of adjacent rows of battery cells **10** in the second direction Y have non-overlapping portions **20**.

[0123] As shown in FIG. 1, FIG. 2 and FIG. 4, the heat conducting member **40** is arranged in the box body **30** and is constructed to be deformable to buffer a stress. The heat conducting member **40** includes a heat conducting layer **41** and a heat insulating layer **42**. The heat conducting layer **41** and the heat insulating layer **42** are arranged in a stacked manner, and a ratio of the thickness of the heat insulating layer **42** to the thickness of the heat conducting layer **41** is less than $\frac{1}{2}$.

[0124] As shown in FIG. 1 and FIG. 2, the heat conducting member **40** is located between the second side walls **112** of each row of the battery units **10A** and the box body **30**, so that the heat conducting layer **41** of the heat conducting member **40** is connected to the second side walls **112** of the battery cells **10** at end portions of each row of the battery units **10A** in a heat conducting manner, and the heat insulating layer **42** is in contact with the non-overlapping portions **20** in a heat insulating manner.

[0125] A first heat conducting adhesive layer is provided between the heat conducting layer **41** and bottom and side walls of the box body **30**. A second heat conducting adhesive layer is provided between the heat conducting layer **41** and the opposite second side wall **112**.

[0126] A power consuming apparatus **1000** according to an embodiment in a second aspect of the present application will be described below with reference to the drawings.

[0127] As shown in FIG. 5, the power consuming apparatus **1000** according to the embodiment of the present application includes the battery **200** described in the above embodiment. The battery **200** is configured to provide electric energy.

[0128] Since the battery **200** in the embodiment of the present application has the above technical effects, the power consuming apparatus **1000** in the embodiment of the present application also has the above technical effects, that is, by using the battery **200** in the present application, the use safety of the power consuming apparatus **1000** can be improved, and the service life of the power consuming apparatus **1000** can be extended.

[0129] It should be noted that according to an embodiment in a second aspect of the present application, a power consuming apparatus **1000** using the battery **200** as a power supply is provided. The power consuming apparatus **1000** may be, but not limited to, a mobile phone, a tablet, a laptop, an electric toy, a power tool, an electromobile, an electric vehicle, a vehicle, a ship, a spacecraft, or the like. The electric toy may include a fixed or mobile electric toy, such as a game console, an electric vehicle toy, an electric ship toy, an electric airplane toy, or the like. The spacecraft may include an airplane, a rocket, a space shuttle, a spacecraft, or the like.

[0130] FIG. 5 shows a schematic diagram when the power consuming apparatus **1000** is a vehicle. It should be noted that when the power consuming apparatus **1000** is a vehicle, the vehicle may be an oil-fueled vehicle, a gas vehicle, or a new energy vehicle. The new energy vehicle may be a pure electric vehicle, a hybrid vehicle, an extended range vehicle, or the like.

[0131] As shown in FIG. 5, the battery **200** is arranged in the vehicle. The battery **200** may be arranged at a bottom, a head or a tail of the vehicle. The battery **200** here may be configured to supply power to the vehicle. For example, the battery **200** may serve as an operating power supply of the vehicle. The vehicle may further include a controller **300** and a motor **400**. The controller **300** is configured to control the battery **200** to supply power to the motor **400**, for example, to meet the working power demand during vehicle start-up, navigation, and traveling.

[0132] In some embodiments of the present application, the battery **200** may not only serve as an operating power supply of the vehicle, but also serve as a driving power supply of the vehicle to replace or partially replace fuel or natural gas to provide driving power for the vehicle.

[0133] It may be understood that other compositions and operations of the battery **200** and the power consuming apparatus **1000** with the same according to the embodiments of the present

application are known to those of ordinary skill in the art, which will not be described here in detail.

[0134] It should be noted that without conflict, the embodiments and features in the embodiments of the present application may be combined with each other.

[0135] What are described above are just preferred embodiments of the present application and are not intended to limit the present application. For those skilled in the art, the present application may have various modifications and variations. Any modification, equivalent replacement, improvement, and the like made within the spirit and principle of the present application should be included in the scope of protection of the present application.

Claims

1. A battery, comprising: a box body; a plurality of rows of battery units, wherein the plurality of rows of battery units are arranged in the box body, each row of the battery units comprises a plurality of battery cells arranged along a first direction, each battery cell is provided with a plurality of side walls, the plurality of side walls comprise a first side wall and a second side wall, the first side wall and the second side wall are connected to each other, the first side wall is a side wall with the largest area of the battery cell, at least two rows of the battery units are arranged in a staggered manner in the first direction so that the first side walls of adjacent rows of battery cells arranged opposite each other are staggered in the first direction; and a heat conducting member, wherein the heat conducting member is arranged in the box body, the heat conducting member is provided between the second side walls of at least one row of the battery units and the box body in the first direction so that the second side walls of at least part of the battery cells are connected to the box body in a heat conducting manner through the heat conducting member.
 2. The battery according to claim 1, wherein the heat conducting member is connected to different side walls of the box body in a heat conducting manner.
 3. The battery according to claim 2, wherein a first heat conducting adhesive layer is provided between the heat conducting member and the box body.
 4. The battery according to claim 1, wherein orthographic projections of the first side walls of the adjacent rows of battery cells arranged in a staggered manner in a second direction have non-overlapping portions, and the non-overlapping portions are in contact with the heat conducting member in a heat insulating manner.
 5. The battery according to claim 4, wherein the heat conducting member comprises a heat conducting layer and a heat insulating layer arranged in a stacked manner, the heat conducting layer is connected to the second side wall in a heat conducting manner, and the heat insulating layer is in contact with the non-overlapping portions.
 6. The battery according to claim 5, wherein a second heat conducting adhesive layer is provided between the heat conducting layer and the opposite second side wall.
 7. The battery according to claim 5, wherein a thickness of the heat conducting layer is greater than that of the heat insulating layer.
 8. The battery according to claim 7, wherein a ratio of the thickness of the heat insulating layer to the thickness of the heat conducting layer is not greater than $\frac{1}{2}$.
 9. The battery according to claim 1, wherein the heat conducting member is filled in a gap between the battery cells and an inner wall of the box body.
 10. The battery according to claim 1, wherein the heat conducting member is constructed to be deformable to buffer a stress.
 11. A power consuming apparatus, comprising the battery according to claim 1, the battery being configured to provide electric energy.
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